

Qualification Pack



Soil and Water Conservation Supervisor

QP Code: CUTM/AGR/Q1301

Version: 1.0

NSQF Level: 5

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CUTM/AGR/Q1301: Soil and Water Conservation Supervisor

Brief Job Description

The individual will get detailed insights on watershed processes and hydro-climatic data processing technique, scientific methodology of maintaining data records, maintenance of equipment, design consideration of different soil and water conservation structures for minimizing soil erosion and improving crop production.

Personal Attributes

A Soil and Water Conservation Supervisor should be environmentally conscious, responsible, and committed to sustainable resource management. They must possess strong observation and analytical skills to monitor soil health, water flow, and conservation measures. Leadership and decision-making abilities are essential for guiding field teams and implementing projects effectively. The role requires adaptability to diverse terrains and climatic conditions, along with problem-solving skills for addressing erosion, water scarcity, and land degradation. Good communication and planning skills are important for coordinating with farmers, communities, and government agencies. Patience, attention to detail, and dedication to environmental protection are key personal attributes.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [CUTM/AGR/N1301: Understand About Various Watersheds Processes](#)
2. [CUTM/AGR/N1302: Learn Data acquisition and processing techniques for watershed management](#)
3. [CUTM/AGR/N1303: Perform the process of delineation of watershed](#)
4. [CUTM/AGR/N1304: Study About the Processes to Improve Soil Quality](#)
5. [CUTM/AGR/N1305: Design of different Soil and Water Conservation Structures based requirements](#)
6. [CUTM/AGR/N1306: Improve Groundwater Quality and Recharge](#)
7. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Agriculture
Sub-Sector	Sustainable Agriculture

Qualification Pack

Occupation	Soil Conservationist
Country	India
NSQF Level	5
Credits	21
Aligned to NCO/ISCO/ISIC Code	NSQF
Minimum Educational Qualification & Experience	Completed 2nd year of UG (UG Diploma) (in agriculture/agricultural engineering) with NA of experience OR Completed 1st year of UG (UG Certificate) (in agriculture/agricultural engineering) with 1.5 years of experience relevant Experience OR 12th grade Pass with 3 Years of experience relevant Experience OR Previous relevant Qualification of NSQF Level (4) with 3 Years of experience relevant Experience OR Previous relevant Qualification of NSQF Level (4.5) with 1.5 years of experience relevant Experience
Minimum Level of Education for Training in School	12th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	08/05/2028
NSQF Approval Date	08/05/2025
Version	1.0
Reference code on NQR	QG-05-AG-04286-2025-V1-CUTM
NQR Version	1.0

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CUTM/AGR/N1301: Understand About Various Watersheds Processes

Description

This NOS focuses on imparting knowledge of watershed dynamics and their role in soil and water conservation. Learners will study hydrological cycles, soil erosion patterns, land-use interactions, and the significance of groundwater recharge. It covers collection and interpretation of agricultural, climatic, and environmental data, as well as identification of critical watershed issues. The NOS also emphasizes prioritization of watersheds, problem analysis, and preparation of evaluation reports to support sustainable land and water resource management.

Scope

The scope covers the following :

- This NOS focuses on developing knowledge and skills to understand the functioning of watershed systems and their impact on soil and water resources. It covers collection of localized information, hydrological and climatic data, agricultural practices, and interpretation of topographic and geological maps. Learners will identify critical watershed issues, such as soil erosion hazards and groundwater recharge potential, and prioritize watersheds accordingly. The scope also includes preparing evaluation reports that support sustainable watershed management and conservation practices.

Elements and Performance Criteria

Understand about various watersheds processes

To be competent, the user/individual on the job must be able to:

- PC1.** Collection of localized information related to selected watershed.
- PC2.** Understand various processes involved in a watershed.
- PC3.** Collect agricultural practice information, climatic data, hydrological data, environmental data, aerial photos, topographic maps, and geological maps from relevant sources.
- PC4.** Identify critical problems of various watersheds based on review of literatures.
- PC5.** Prioritization of Watersheds based on soil erosion hazard and quality of groundwater recharge.
- PC6.** Make a report based on critical evaluation of the watershed.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Concept of Watersheds - Understanding the definition, components, and importance of a watershed, including its role in soil and water conservation, agriculture, and ecosystem balance.
- KU2.** Hydrological Cycle & Processes - Knowledge of precipitation, infiltration, runoff, evapotranspiration, and groundwater recharge, and how these processes interact within a watershed.

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- KU3.** Data Collection for Watershed Analysis - Familiarity with sources and methods of collecting agricultural, climatic, hydrological, environmental, and geological data, including use of maps, aerial photos, and meteorological records.
- KU4.** Watershed Characterization - Understanding land use, soil types, topography, vegetation cover, and socio-economic conditions that influence watershed behavior.
- KU5.** Critical Problem Identification - Ability to identify and analyze issues such as soil erosion, sedimentation, flooding, declining groundwater levels, and pollution in watersheds.
- KU6.** Watershed Prioritization - Knowledge of methods to prioritize watersheds based on erosion hazards, sediment yield, groundwater recharge potential, and environmental concerns.
- KU7.** Watershed Delineation - Understanding techniques for delineating watershed boundaries using topographic features, Digital Elevation Models (DEM), and GIS tools.
- KU8.** Soil Erosion and Sediment Transport - Awareness of erosion processes (sheet, rill, gully) and their impact on watershed sustainability and agricultural productivity.
- KU9.** Evaluation & Reporting - Skills in preparing scientific reports based on watershed data analysis, including interpretation of hydro-meteorological and environmental findings.
- KU10.** Sustainability & Conservation Linkages - Understanding how watershed management contributes to soil fertility, water availability, climate resilience, and sustainable agricultural practices.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** explain the definition, components, and functions of a watershed, including hydrological cycle and linkages between land, soil, and water systems.
- GS2.** collect localized information such as agricultural practices, climatic data, hydrological data, environmental parameters, aerial photos, and topographic maps for watershed studies.
- GS3.** identify and assess critical watershed problems like soil erosion, runoff intensity, groundwater depletion, and waterlogging, using literature reviews and field observations.
- GS4.** prioritize watersheds based on soil erosion hazards, water recharge potential, and socio-economic importance for effective conservation planning.
- GS5.** conduct ground-truth surveys, site inspections, and verification of secondary data to ensure accuracy in watershed studies.
- GS6.** prepare structured technical reports, integrating scientific data and field findings to support watershed management decisions.
- GS7.** Use the Scientific Tools & Techniques - Familiarity with basic watershed-related instruments and analysis methods (e.g., soil augers, hydrometers, pH meters, GIS-based mapping tools).
- GS8.** evaluate multiple solutions for watershed-related challenges and adopt the most sustainable and cost-effective measures.
- GS9.** communicate watershed-related findings clearly with farmers, local communities, and government agencies, and to collaborate with multidisciplinary teams.
- GS10.** understand watershed processes contribute to sustainable agriculture, soil conservation, water resource management, and climate resilience.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understand about various watersheds processes</i>	40	50	-	10
PC1. Collection of localized information related to selected watershed.	1	10	-	2
PC2. Understand various processes involved in a watershed.	7	10	-	2
PC3. Collect agricultural practice information, climatic data, hydrological data, environmental data, aerial photos, topographic maps, and geological maps from relevant sources.	8	10	-	2
PC4. Identify critical problems of various watersheds based on review of literatures.	8	10	-	2
PC5. Prioritization of Watersheds based on soil erosion hazard and quality of groundwater recharge.	8	10	-	2
PC6. Make a report based on critical evaluation of the watershed.	8	-	-	-
NOS Total	40	50	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N1301
NOS Name	Understand About Various Watersheds Processes
Sector	Agriculture
Sub-Sector	
Occupation	Soil Conservationist
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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CUTM/AGR/N1302: Learn Data acquisition and processing techniques for watershed management

Description

This NOS focuses on equipping learners with skills to systematically collect, analyze, and process data critical for effective watershed management. It covers meteorological, soil, and satellite data collection, ground-truthing, and the use of GIS/remote sensing tools for trend analysis and land use change detection. Learners will also gain proficiency in software such as QGIS and ArcGIS to process and interpret data. The outcome ensures the ability to prepare actionable reports for farmers and stakeholders, enabling informed decision-making for sustainable soil and water conservation practices.

Scope

The scope covers the following :

- The scope includes gathering field data from local and regional sources, performing soil sampling and analysis, using GIS and remote sensing tools (QGIS, ArcGIS) for change detection and land-use studies, and preparing analytical reports. This NOS ensures supervisors can integrate scientific data with practical watershed management strategies, supporting sustainable soil and water conservation practices.

Elements and Performance Criteria

Data Acquisition and Processing Techniques for Watershed Management

To be competent, the user/individual on the job must be able to:

- PC1.** Collect meteorological data at local scale from nearby meteorological laboratory (e.g. Agricultural University, Govt. agricultural dept., Govt. Irrigation dept. etc.)
- PC2.** Trend analysis of collected meteorological data.
- PC3.** Collect soil data at local scale by sampling technique.
- PC4.** Soil sample analysis (mention if any assumption made regarding grid size).
- PC5.** Collect soil data at regional scale from National Bureau of Soil Survey & Land use Planning.
- PC6.** Collect satellite data from USGS Earth explorer website.
- PC7.** Change detection analysis of land use land cover of the watershed.
- PC8.** Process the collected satellite data by using software like QGIS and ArcGIS etc.
- PC9.** Make a report based on critical assessment and suggest farmers to act on that.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** methods to collect local-scale meteorological data (rainfall, temperature, humidity, wind speed, etc.) from relevant government or agricultural meteorological stations.
- KU2.** how to perform trend analysis on collected hydro-meteorological data to understand climate variability, rainfall intensity patterns, and seasonal changes.

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- KU3.** soil sampling techniques at local scale and grid-based methods to ensure representative data collection on soil type, texture, fertility, and moisture.
- KU4.** how to conduct laboratory-based soil analysis for pH, nutrients, moisture, and physical properties, as well as collecting secondary soil data from institutions like the National Bureau of Soil Survey & Land Use Planning.
- KU5.** downloading and using satellite data (e.g., from USGS Earth Explorer) for watershed-level monitoring, mapping, and change detection.
- KU6.** how to process and analyze spatial datasets using GIS software like QGIS and ArcGIS, including handling thematic layers, land use/land cover classification, and watershed mapping.
- KU7.** change detection techniques for monitoring land use/land cover changes, soil erosion patterns, and vegetation cover over time in watershed areas.
- KU8.** various field instruments used for collecting hydro-meteorological and soil data, ensuring accuracy and validation of records.
- KU9.** how to integrate meteorological, soil, hydrological, environmental, and remote sensing data for holistic watershed management planning.
- KU10.** identifying key watershed issues, and suggesting actionable strategies for farmers and local communities to improve resource management.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** collect climate-related data such as rainfall, temperature, humidity, and wind speed from local sources (universities, agriculture/irrigation departments, etc.), ensuring reliability for watershed analysis.
- GS2.** conduct soil sampling at local and regional levels using proper techniques, followed by laboratory analysis to determine soil texture, fertility, pH, and nutrient status.
- GS3.** gather hydrological records, monitoring water quality parameters, and integrating them into watershed management frameworks.
- GS4.** acquire geospatial datasets (e.g., from USGS Earth Explorer) and apply remote sensing techniques for watershed study.
- GS5.** use GIS software such as QGIS and ArcGIS for processing spatial data, conducting land use/land cover change detection, and preparing thematic maps.
- GS6.** perform time-series and trend analysis of meteorological and hydrological data, and identify land use/land cover changes within watershed boundaries.
- GS7.** prepare technical reports that synthesize soil, water, and climate data analyzing outcomes and suggesting evidence-based interventions to farmers and local authorities.
- GS8.** apply statistical techniques for data validation, correlation of variables, and deriving meaningful insights for watershed prioritization.
- GS9.** create maps/graphs, and communicate results effectively to stakeholders including farmers, policymakers, and community members.
- GS10.** Use acquired data to support decision-making in watershed management, advising on soil conservation practices, crop planning, and sustainable resource use.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Data Acquisition and Processing Techniques for Watershed Management</i>	40	50	-	-
PC1. Collect meteorological data at local scale from nearby meteorological laboratory (e.g. Agricultural University, Govt. agricultural dept., Govt. Irrigation dept. etc.)	5	10	-	-
PC2. Trend analysis of collected meteorological data.	5	10	-	-
PC3. Collect soil data at local scale by sampling technique.	5	5	-	-
PC4. Soil sample analysis (mention if any assumption made regarding grid size).	5	5	-	-
PC5. Collect soil data at regional scale from National Bureau of Soil Survey & Land use Planning.	5	5	-	-
PC6. Collect satellite data from USGS Earth explorer website.	5	5	-	-
PC7. Change detection analysis of land use land cover of the watershed.	5	5	-	-
PC8. Process the collected satellite data by using software like QGIS and ArcGIS etc.	5	5	-	-
PC9. Make a report based on critical assessment and suggest farmers to act on that.	-	-	-	-
NOS Total	40	50	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N1302
NOS Name	Learn Data acquisition and processing techniques for watershed management
Sector	Agriculture
Sub-Sector	
Occupation	Soil Conservationist
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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CUTM/AGR/N1303: Perform the process of delineation of watershed

Description

This NOS involves applying scientific and geospatial techniques to delineate watershed boundaries accurately. It covers the use of Digital Elevation Models (DEM), hydro-meteorological data, and GIS tools to analyze flow direction, accumulation, and stream order. The individual is expected to interpret maps, conduct soil and land use analysis, and prepare thematic layers for effective watershed planning. By performing delineation, the supervisor ensures better management of soil and water resources, enabling sustainable agricultural practices and improved conservation outcomes.

Scope

The scope covers the following :

- It covers the study of hydro-meteorological data, application of Digital Elevation Models (DEM), use of GIS and remote sensing techniques, and interpretation of map features such as flow direction, accumulation, and stream order. This scope ensures accurate watershed delineation, which is critical for effective planning, resource management, and implementation of soil and water conservation measures.

Elements and Performance Criteria

Delineation of Watershed

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the process of Watershed delineation.
- PC2.** Learn Digital Elevation Model data.
- PC3.** Be familiar with various processes involved in watershed delineation.
- PC4.** Study about various map features (flow direction, flow accumulation, stream order etc.)

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** what watershed delineation means, its importance in hydrology, and its role in soil and water conservation planning.
- KU2.** Digital Elevation Model (DEM) Data - Knowledge of DEMs, their sources (e.g., satellite imagery), and how elevation data is used to define watershed boundaries.
- KU3.** Processes in Watershed Delineation - Familiarity with steps such as identifying ridgelines, drainage patterns, catchment areas, and defining watershed outlets.
- KU4.** what watershed delineation means, its importance in hydrology, and its role in soil and water conservation planning.
- KU5.** what watershed delineation means, its importance in hydrology, and its role in soil and water conservation planning.
- KU6.** what watershed delineation means, its importance in hydrology, and its role in soil and water conservation planning.

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- KU7.** what watershed delineation means, its importance in hydrology, and its role in soil and water conservation planning.
- KU8.** what watershed delineation means, its importance in hydrology, and its role in soil and water conservation planning.
- KU9.** what watershed delineation means, its importance in hydrology, and its role in soil and water conservation planning.
- KU10.** what watershed delineation means, its importance in hydrology, and its role in soil and water conservation planning.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** explain watershed boundaries, hydrological units, and their role in soil and water conservation planning.
- GS2.** obtain and use DEM data to analyze topography, slope, and drainage networks for delineation tasks.
- GS3.** use software such as QGIS/ArcGIS to generate thematic layers, analyze flow direction, flow accumulation, and stream order for watershed mapping.
- GS4.** interpret rainfall, runoff, and climatic data for assessing watershed behavior and prioritization.
- GS5.** use soil sampling techniques and laboratory analysis to support delineation decisions, including nutrient availability and erosion risk.
- GS6.** use skill in analyzing and interpreting topographic maps, satellite images, and aerial photos preparing clear reports with delineated watershed boundaries.
- GS7.** evaluate multiple watershed management approaches, identify suitable conservation measures, and recommend farmer-friendly interventions.
- GS8.** to use hydrological and soil testing equipment (e.g., hydrometers, soil augers, pH meters) while ensuring accuracy and safety.
- GS9.** communicate technical findings effectively with farmers, local communities, and supervisors through reports, maps, and presentations.
- GS10.** Ensure delineation outcomes support soil conservation, groundwater recharge, and sustainable land-use planning in line with environmental goals.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Delineation of Watershed</i>	40	50	-	-
PC1. Understand the process of Watershed delineation.	-	-	-	-
PC2. Learn Digital Elevation Model data.	10	20	-	-
PC3. Be familiar with various processes involved in watershed delineation.	20	10	-	-
PC4. Study about various map features (flow direction, flow accumulation, stream order etc.)	10	20	-	-
NOS Total	40	50	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N1303
NOS Name	Perform the process of delineation of watershed
Sector	Agriculture
Sub-Sector	
Occupation	Soil Conservationist
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQF Clearance Date	08/05/2025

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CUTM/AGR/N1304: Study About the Processes to Improve Soil Quality

Description

This NOS focuses on enhancing and maintaining soil health to ensure sustainable agricultural productivity. It covers techniques to increase soil organic matter, monitor and balance nutrient levels, manage soil pH, and improve soil structure through aggregation and porosity management. Trainees will also learn methods to assess microbial activity, control soil erosion, and track contaminant levels. By applying these practices, soil quality can be improved for better water retention, crop yield, and long-term ecological balance.

Scope

The scope covers the following :

- It covers improving soil organic matter, maintaining optimal pH and nutrient levels, assessing soil structure (aggregation, porosity, compaction), monitoring erosion and sediment loss, and evaluating microbial activity. The scope also includes identifying soil contaminants, applying soil amendments, and adopting organic and conservation practices. These processes help improve fertility, crop productivity, and long-term soil sustainability, supporting effective watershed and land management.

Elements and Performance Criteria

Improve Soil Quality

To be competent, the user/individual on the job must be able to:

- PC1.** Improve Soil Organic Matter (SOM) Content.
- PC2.** Monitor and improve nutrient levels.
- PC3.** Maintain optimal soil pH levels.
- PC4.** Assessing soil aggregation, porosity, and compaction by implementing improved soil structure.
- PC5.** Monitor soil erosion rates and sediment loss.
- PC6.** Estimate microbial biomass, enzyme activity, and other indicators of soil health.
- PC7.** Monitor soil contaminant levels over time.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the role of organic matter in soil fertility, its sources (crop residues, compost, green manure), and factors affecting its decomposition.
- KU2.** Nutrient Management: Knowledge of essential soil nutrients (N, P, K, micro-nutrients), their functions, and methods of monitoring and balancing nutrient levels for healthy crop production.
- KU3.** the importance of maintaining optimal soil pH, use of liming materials or amendments, and impacts of acidic or alkaline soils on crop growth.

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- KU4.** about soil aggregation, porosity, permeability, and compaction, and how these properties affect water retention, aeration, and root growth.
- KU5.** Soil Erosion and Sedimentation: Awareness of soil erosion processes, methods to monitor soil erosion rates, and the use of GIS and field-scale measurement techniques to quantify soil loss.
- KU6.** the role of microorganisms, microbial biomass, and enzyme activity in maintaining soil health and nutrient cycling.
- KU7.** Soil Contamination and Pollution Monitoring: Knowledge of sources of soil contamination (pesticides, heavy metals, industrial waste) and methods to monitor and control soil contaminant levels over time.
- KU8.** sustainable practices such as crop rotation, cover cropping, green manures, composting, organic fertilizers, mulching, conservation tillage, and livestock-based soil fertility improvement.
- KU9.** indicators of soil health (chemical, physical, and biological), including EC (Electrical Conductivity), organic matter content, and microbial counts.
- KU10.** the use of modern tools and techniques such as soil testing kits, spectrophotometers, GIS-based soil monitoring, and data analysis for effective soil quality improvement strategies.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** increase organic matter in soil using compost, green manures, cover crops, crop residues, and livestock-based practices (e.g., mob grazing). This improves soil fertility and water-holding capacity.
- GS2.** analyze and balance nutrient levels (N, P, K, micronutrients) through soil testing and application of organic or inorganic amendments, ensuring crop productivity.
- GS3.** measure and adjust soil pH using lime or sulfur-based materials. Maintaining optimal pH enhances nutrient availability and crop growth.
- GS4.** assess aggregation, porosity, and compaction levels, and implement practices (reduced tillage, organic inputs, mulching) that improve soil structure and aeration.
- GS5.** Monitor Erosion and Sedimentation : Competence in field-scale measurement of erosion, quantification of sediment loss using GIS techniques, and recommending preventive measures like contour farming and vegetative barriers.
- GS6.** Conduct microbial biomass estimation, enzyme activity analysis, and microbial counts to evaluate soil biological health and enhance ecosystem functions.
- GS7.** monitor soil pollutant levels (heavy metals, pesticides, industrial residues) over time, ensuring soil safety for agriculture and the environment.
- GS8.** Use of GIS mapping, soil moisture sensors, and electrical conductivity meters to quantify and track soil quality parameters at field and watershed levels.
- GS9.** Adopt Sustainable Practices: Professional application of conservation agriculture, crop rotation, agroforestry, and organic farming to maintain long-term soil health and prevent degradation.
- GS10.** Prepare technical reports on soil quality assessments and effectively guide farmers/communities with scientific solutions to improve soil fertility, ensuring both productivity and environmental sustainability.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Improve Soil Quality</i>	40	50	10	-
PC1. Improve Soil Organic Matter (SOM) Content.	5	5	-	-
PC2. Monitor and improve nutrient levels.	5	5	-	-
PC3. Maintain optimal soil pH levels.	5	10	-	-
PC4. Assessing soil aggregation, porosity, and compaction by implementing improved soil structure.	5	10	10	-
PC5. Monitor soil erosion rates and sediment loss.	5	5	-	-
PC6. Estimate microbial biomass, enzyme activity, and other indicators of soil health.	10	10	-	-
PC7. Monitor soil contaminant levels over time.	5	5	-	-
NOS Total	40	50	10	-

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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N1304
NOS Name	Study About the Processes to Improve Soil Quality
Sector	Agriculture
Sub-Sector	
Occupation	Soil Conservationist
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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CUTM/AGR/N1305: Design of different Soil and Water Conservation Structures based requirements

Description

This NOS focuses on planning and designing soil and water conservation structures to minimize erosion, retain soil fertility, and enhance sustainable land use. It covers agricultural measures such as contour farming, crop rotation, strip cropping, mulching, and organic farming, agroforestry practices like agri-silviculture and alley cropping, and mechanical measures including contour bunds, graded bunds, trenches, terraces, and check dams. Learners gain knowledge to select appropriate techniques based on site conditions, ensuring effective soil and water management for long-term agricultural productivity.

Scope

The scope covers the following :

- This NOS covers the knowledge and skills required to design soil and water conservation structures by integrating agricultural, agroforestry, and mechanical measures. It includes contour farming, crop rotation, strip cropping, mulching, conservation tillage, and organic farming practices to minimize soil erosion. The scope also extends to agroforestry systems like agri-silviculture and alley cropping, along with mechanical interventions such as contour bunds, graded bunds, trenches, terraces, and check dams. The focus is on selecting site-specific, sustainable solutions to conserve soil, enhance water retention, and support long-term agricultural productivity.

Elements and Performance Criteria

Agricultural Measures for soil and water conservation

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the designing procedure of contour Farming.
- PC2.** Select suitable crops to control soil and water loss.
- PC3.** Study the importance of crop rotation for soil conservation.
- PC4.** Learn cover crops method.
- PC5.** Understand intercropping system.
- PC6.** Study contour strip cropping.
- PC7.** Learn field strip cropping.
- PC8.** Study wind strip cropping.
- PC9.** Understand buffer strip cropping.
- PC10.** Learn plastic Mulching.
- PC11.** Understand conservation tillage.
- PC12.** Learn adoption of organic farming strategies for controlling soil loss.
- PC13.** Study ridge and furrow farming system of farming.
- PC14.** Understand broad bed and furrow system of farming.

Agroforestry measures for soil and water conservation

To be competent, the user/individual on the job must be able to:

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- PC15.** Learn Agri – silvi culture.
- PC16.** Study Agri-Horticulture.
- PC17.** Understand Alley cropping.
- PC18.** Learn Silvi-pasture system.

Mechanical measures for soil and water conservation

To be competent, the user/individual on the job must be able to:

- PC19.** Design Contour bunding.
- PC20.** Design Graded bunding.
- PC21.** Learn Peripheral bunding.
- PC22.** Design continuous contour trenches.
- PC23.** Design staggered contour trenches.
- PC24.** Learn designing procedure of Bench terraces.
- PC25.** Understand designing procedure of loose boulder/stone/masonry check dams.
- PC26.** Understand designing procedure of brushwood check dams.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the design and application of contour farming, crop rotation, intercropping, strip cropping (contour, field, wind, buffer), cover cropping, mulching, conservation tillage, ridge & furrow system, and broad bed & furrow farming for soil and water conservation.
- KU2.** Crop Selection for Conservation - Knowledge of selecting suitable crops and cropping systems that reduce soil erosion and water loss, while improving land productivity and sustainability.
- KU3.** agroforestry measures such as Agri-silviculture, Agri-horticulture, Alley cropping, and Silvi-pasture systems that integrate trees and crops to conserve soil and water.
- KU4.** Mechanical Conservation Structures - Knowledge of designing contour bunding, graded bunding, peripheral bunding, continuous and staggered contour trenches, bench terraces, check dams (loose boulder/stone/masonry), and brushwood dams for effective erosion control and water management.
- KU5.** the design principles, layout methods, dimensioning, and slope calculations involved in planning different conservation structures.
- KU6.** processes like runoff, infiltration, evaporation, and erosion that determine the necessity and type of conservation structures to be adopted.
- KU7.** Integration of Conservation Measures - Knowledge of combining agronomic, agroforestry, and mechanical methods into a watershed management plan tailored to local topography, soil type, and rainfall patterns.
- KU8.** organic farming approaches and soil health practices that complement physical structures to reduce soil degradation.
- KU9.** factors such as land slope, rainfall intensity, soil texture, vegetation cover, and farmer practices in determining the right structure for a given location.
- KU10.** how to apply knowledge in designing, demonstrating, and supervising construction/implementation of soil and water conservation structures in real field condition

Qualification Pack

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** acquire knowledge of agricultural practices such as contour farming, crop rotation, strip cropping, mulching, and conservation tillage to minimize soil erosion and enhance soil productivity.
- GS2.** identify suitable crops for controlling water and soil loss, implementing intercropping, ridge - furrow, and broad-bed - furrow systems based on land and climate requirements.
- GS3.** apply organic farming, cover cropping, and agroforestry methods (agri-silviculture, agri-horticulture, silvi-pasture) to promote soil conservation and long-term sustainability.
- GS4.** design and evaluate mechanical conservation structures like contour bunds, graded bunds, peripheral bunds, continuous/staggered contour trenches, and bench terraces, tailored to topography.
- GS5.** design small-scale check dams (loose boulder, stone masonry, brushwood) to regulate water flow, reduce gully erosion, and improve groundwater recharge.
- GS6.** incorporate tree-based systems (alley cropping, wind strip cropping, buffer strips) that integrate soil conservation with livelihood and watershed management needs.
- GS7.** follow step-by-step technical design calculations, prepare layout plans, and apply GIS/maps for structure placement as per soil and water conservation engineering standards.
- GS8.** to develop practical problem-solving skills by assessing field conditions, soil erosion intensity, slope, and hydrological factors before selecting and designing conservation structures.
- GS9.** adhere to safety protocols while constructing structures and ensuring eco-friendly practices that align with environmental sustainability goals.
- GS10.** do document designs, prepare technical reports, and effectively communicate with farmers, field workers, and supervisors to implement soil and water conservation projects.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Agricultural Measures for soil and water conservation</i>	25	37	-	-
PC1. Understand the designing procedure of contour Farming.	2	2	-	-
PC2. Select suitable crops to control soil and water loss.	3	3	-	-
PC3. Study the importance of crop rotation for soil conservation.	2	3	-	-
PC4. Learn cover crops method.	2	3	-	-
PC5. Understand intercropping system.	2	3	-	-
PC6. Study contour strip cropping.	2	3	-	-
PC7. Learn field strip cropping.	3	3	-	-
PC8. Study wind strip cropping.	3	3	-	-
PC9. Understand buffer strip cropping.	1	2	-	-
PC10. Learn plastic Mulching.	1	3	-	-
PC11. Understand conservation tillage.	1	2	-	-
PC12. Learn adoption of organic farming strategies for controlling soil loss.	1	3	-	-
PC13. Study ridge and furrow farming system of farming.	1	3	-	-
PC14. Understand broad bed and furrow system of farming.	1	1	-	-
<i>Agroforestry measures for soil and water conservation</i>	4	4	5	-
PC15. Learn Agri – silvi culture.	1	1	-	-
PC16. Study Agri-Horticulture.	1	1	-	-
PC17. Understand Alley cropping.	1	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC18. Learn Silvi-pasture system.	1	1	-	-
<i>Mechanical measures for soil and water conservation</i>	11	9	5	-
PC19. Design Contour bunding.	1	1	-	-
PC20. Design Graded bunding.	1	1	-	-
PC21. Learn Peripheral bunding.	1	1	-	-
PC22. Design continuous contour trenches.	3	1	-	-
PC23. Design staggered contour trenches.	2	2	-	-
PC24. Learn designing procedure of Bench terraces.	1	1	-	-
PC25. Understand designing procedure of loose boulder/stone/masonry check dams.	1	1	-	-
PC26. Understand designing procedure of brushwood check dams.	1	1	-	-
NOS Total	40	50	10	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N1305
NOS Name	Design of different Soil and Water Conservation Structures based requirements
Sector	Agriculture
Sub-Sector	
Occupation	Soil Conservationist
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

CUTM/AGR/N1306: Improve Groundwater Quality and Recharge

Description

This NOS focuses on enhancing groundwater sustainability through quality monitoring and recharge interventions. It covers reducing contaminants such as nitrates, pesticides, and heavy metals, and enforcing protective measures for recharge areas. Learners gain skills in assessing groundwater toxicity, implementing stormwater management, artificial recharge techniques, and improving soil permeability. Emphasis is placed on hydrological modeling, use of monitoring wells, community engagement, and conservation practices to maintain aquifer health and ensure sustainable water availability.

Scope

The scope covers the following :

- It covers monitoring and reducing contaminants like nitrates, pesticides, and heavy metals, ensuring safe water for human and ecological health. The scope includes assessing recharge rates through hydrological modeling, adopting artificial recharge methods (recharge wells, infiltration basins), Storm water management, and soil improvement practices to enhance infiltration. It also emphasizes wetland restoration, protected recharge zones, water conservation, and community engagement for long-term groundwater sustainability.

Elements and Performance Criteria

Groundwater Quality Improvement

To be competent, the user/individual on the job must be able to:

- PC1.** Monitor and reduce the concentration of contaminants such as nitrates, pesticides, heavy metals, and other pollutants in groundwater.
- PC2.** Assess toxicity of groundwater to aquatic organisms and human health through bioassays and chemical analyses.
- PC3.** Implement groundwater extraction practices that do not deplete aquifer resources beyond recharge rates, prevent saltwater intrusion and land subsidence.
- PC4.** Establish and enforce regulations to protect recharge areas from contamination sources, such as industrial sites, landfills, and agricultural runoff.

Groundwater Recharge Improvement

To be competent, the user/individual on the job must be able to:

- PC5.** Monitor changes in groundwater recharge rates through hydrological modeling, groundwater level measurements, and recharge area assessments
- PC6.** Implement storm water management practices (e.g., rain gardens, permeable pavement) to enhance infiltration and groundwater recharge from urban areas.
- PC7.** Implement artificial recharge techniques such as infiltration basins, recharge wells, and injection of treated wastewater to augment natural recharge rates.
- PC8.** Improve soil permeability through practices like cover cropping, reduced tillage, and soil amendments to enhance infiltration and recharge rates
- PC9.** Restore or preserve wetlands and riparian zones to promote groundwater recharge through increased groundwater-surface water interaction

Qualification Pack

- PC10.** Establish protected areas or buffer zones around recharge areas to prevent contamination and maintain recharge capacity
- PC11.** Promote water conservation practices to reduce water demand and increase available water for recharge.
- PC12.** Install monitoring wells to track changes in groundwater levels and quality in recharge areas and evaluate the effectiveness of recharge enhancement measures.
- PC13.** Engage local communities in groundwater conservation and recharge efforts through education, outreach, and participation in planning processes

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the sources and impacts of contaminants like nitrates, pesticides, heavy metals, and industrial/agricultural pollutants, and methods to monitor and reduce them.
- KU2.** Toxicity Assessment - Knowledge of bioassays, chemical analyses, and water quality testing to evaluate the toxicity of groundwater for both human health and aquatic ecosystems.
- KU3.** groundwater extraction methods that prevent over-exploitation, saltwater intrusion, and land subsidence while ensuring recharge balance.
- KU4.** regulations, land-use practices, and buffer zones needed to safeguard recharge areas from contamination (industrial waste, landfills, runoff).
- KU5.** hydrological models, groundwater level measurements, and recharge assessments to track recharge effectiveness.
- KU6.** urban water recharge practices such as rain gardens, permeable pavements, and stormwater harvesting to enhance infiltration.
- KU7.** techniques like recharge wells, infiltration basins, and injection of treated wastewater to augment natural recharge processes.
- KU8.** how soil management (cover cropping, reduced tillage, amendments) improves infiltration and facilitates groundwater recharge.
- KU9.** ecological methods for improving recharge through wetland preservation, riparian buffers, and surface - groundwater interaction.
- KU10.** the importance of local community participation, education, and adoption of water conservation practices to sustain recharge efforts.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** test and analyze groundwater for contaminants (nitrates, pesticides, heavy metals, dissolved solids) using modern instruments and field kits.
- GS2.** assess groundwater toxicity impacts on human health and ecosystems and implement safe extraction practices to avoid aquifer depletion, saltwater intrusion, and land subsidence.
- GS3.** follow environmental rules and ability to enforce protective measures around recharge zones to prevent contamination from industrial waste, landfills, and agricultural runoff.

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- GS4.** use hydrological models, GIS tools, and monitoring wells to measure groundwater recharge rates and evaluate water table fluctuations.
- GS5.** design and implement recharge wells, infiltration basins, percolation tanks, and treated wastewater injection methods to enhance aquifer recharge.
- GS6.** Apply agricultural practices like cover cropping, reduced tillage, mulching, and soil amendments to improve infiltration, permeability, and groundwater recharge efficiency.
- GS7.** restore and preserve wetlands, riparian buffers, and vegetative zones to naturally promote recharge through improved groundwater - surface water interaction.
- GS8.** Promote sustainable water use, rainwater harvesting, stormwater management (permeable pavements, rain gardens), and demand-side conservation strategies.
- GS9.** Lead outreach programs to involve local communities, farmers, and stakeholders in groundwater conservation and recharge planning through education and participatory approaches.
- GS10.** Maintain accurate field records, preparing reports on groundwater quality trends, recharge enhancement measures, and presenting findings to decision-makers and farmers for sustainable action.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Groundwater Quality Improvement</i>	6	21	-	-
PC1. Monitor and reduce the concentration of contaminants such as nitrates, pesticides, heavy metals, and other pollutants in groundwater.	2	5	-	-
PC2. Assess toxicity of groundwater to aquatic organisms and human health through bioassays and chemical analyses.	1	6	-	-
PC3. Implement groundwater extraction practices that do not deplete aquifer resources beyond recharge rates, prevent saltwater intrusion and land subsidence.	1	5	-	-
PC4. Establish and enforce regulations to protect recharge areas from contamination sources, such as industrial sites, landfills, and agricultural runoff.	2	5	-	-
<i>Groundwater Recharge Improvement</i>	34	29	10	-
PC5. Monitor changes in groundwater recharge rates through hydrological modeling, groundwater level measurements, and recharge area assessments	3	5	-	-
PC6. Implement storm water management practices (e.g., rain gardens, permeable pavement) to enhance infiltration and groundwater recharge from urban areas.	2	2	-	-
PC7. Implement artificial recharge techniques such as infiltration basins, recharge wells, and injection of treated wastewater to augment natural recharge rates.	3	2	-	-
PC8. Improve soil permeability through practices like cover cropping, reduced tillage, and soil amendments to enhance infiltration and recharge rates	5	2	-	-
PC9. Restore or preserve wetlands and riparian zones to promote groundwater recharge through increased groundwater-surface water interaction	5	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. Establish protected areas or buffer zones around recharge areas to prevent contamination and maintain recharge capacity	5	3	-	-
PC11. Promote water conservation practices to reduce water demand and increase available water for recharge.	5	5	-	-
PC12. Install monitoring wells to track changes in groundwater levels and quality in recharge areas and evaluate the effectiveness of recharge enhancement measures.	3	5	-	-
PC13. Engage local communities in groundwater conservation and recharge efforts through education, outreach, and participation in planning processes	3	3	-	-
NOS Total	40	50	10	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N1306
NOS Name	Improve Groundwater Quality and Recharge
Sector	Agriculture
Sub-Sector	
Occupation	Soil Conservationist
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025



Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Assessment System Overview:

- Batches assigned to the assessment agencies for conducting the assessment on SIDH or email
- Assessment agencies send the assessment confirmation to TP looping AB/AA to accept in SIDH
- Assessment agency deploys the ToA certified Assessor for executing the assessment
- AB monitors the assessment process & records

2. Testing Environment:

- Check the Assessment location, date and time
- If the batch size is more than 30, then there should be 2 Assessors.
- Check that the allotted time to the candidates to complete Theory & Practical Assessment is correct.



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3. Assessment Quality Assurance levels/Framework:

- Question bank is created by the Subject Matter Experts (SME) are verified by the AB
- Questions are mapped to the specified assessment criteria.
- Assessor must be ToA certified & trainer must be ToT Certified

4. Types of evidence or evidence-gathering protocol:

- Time-stamped & geotagged reporting of the assessor from assessment location.
- Centre photographs with signboards and scheme specific branding
- Video Call verification/online proctoring will be done during the assessment from AB.

5. Method of verification or validation:

- Surprise visit to the assessment location

6. Minimum Percentage Required

- 70% overall to pass.
- Minimum 60% in each NOS

7. Reassessment: Candidates failing the assessment will be given one opportunity for reassessment after additional training.

8. Assessor Details: ToA Certified Assessor with relevant experience.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Minimum Passing % at NOS Level: 60

(Please note: A Trainee must score the minimum percentage for each NOS separately as well as on the QP as a

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whole.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CUTM/AGR/N1301.Understand About Various Watersheds Processes	40	50	0	10	100	15
CUTM/AGR/N1302.Learn Data acquisition and processing techniques for watershed management	40	50	0	0	90	15
CUTM/AGR/N1303.Perform the process of delineation of watershed	40	50	0	0	90	15
CUTM/AGR/N1304.Study About the Processes to Improve Soil Quality	40	50	10	0	100	15
CUTM/AGR/N1305.Design of different Soil and Water Conservation Structures based requirements	40	50	10	0	100	15
CUTM/AGR/N1306.Improve Groundwater Quality and Recharge	40	50	10	0	100	15
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	260	330	30	10	630	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.